

Essential Mathematics for Data Science - Quiz 1

Worked solutions (step by step)

Q1. $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$. Compute AB .

Rule: row of A times column of B.

$$\text{Row1} \times \text{Col1} = (1)(2) + (2)(4) = 2 + 8 = 10$$

$$\text{Row1} \times \text{Col2} = (1)(0) + (2)(1) = 0 + 2 = 2$$

$$\text{Row2} \times \text{Col1} = (-1)(2) + (3)(4) = -2 + 12 = 10$$

$$\text{Row2} \times \text{Col2} = (-1)(0) + (3)(1) = 0 + 3 = 3$$

$$AB = \begin{bmatrix} 10 & 2 \\ 10 & 3 \end{bmatrix}$$

Answer: (b) $\begin{bmatrix} 10 & 2 \\ 10 & 3 \end{bmatrix}$

Q2. Find k making $(1,0,1)$, $(0,1,1)$, $(1,k,2)$ linearly dependent.

Dependent means determinant = 0.

$$\det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & k & 2 \end{bmatrix}$$

$$= 1(1 \cdot 2 - 1 \cdot k) - 0(0 \cdot 2 - 1 \cdot 1) + 1(0 \cdot k - 1 \cdot 1)$$

$$= (2 - k) + (-1) = 1 - k$$

$$1 - k = 0 \Rightarrow k = 1$$

Answer: (a) 1

Q3. $\dim \{ x \text{ in } \mathbb{R}^2 \mid x_1 = 2x_2 \}$

Free variable: $x_2 = t$.

Then $x_1 = 2t$, so $x = t(2, 1)$.

One basis vector $(2,1) \Rightarrow 1$ free variable.

Short rule: $\dim = n - (\text{number of independent equations}) = 2 - 1 = 1$

Answer: (b) 1

Q4. $\dim \{ x \text{ in } \mathbb{R}^3 \mid x_1 = 2x_2 + x_3 \}$

Free variables: $x_2 = s$, $x_3 = t$.

$$x = (2s + t, s, t) = s(2, 1, 0) + t(1, 0, 1)$$

Two basis vectors.

$$\text{Rule: } 3 - 1 = 2$$

Answer: (d) 2

Q5. Compute $\begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix}^T$

$$\text{Row1: } (2)(2) + (-1)(-1) = 4 + 1 = 5$$

$$\text{Row2: } (3)(2) + (4)(-1) = 6 - 4 = 2$$

$$\text{Result} = (5, 2)$$

Answer: (c) $(5, 2)$

Q6. Which vector is in span of $(1,0,1)$ and $(0,1,1)$?

General vector: $a(1,0,1) + b(0,1,1) = (a, b, a+b)$.

So the rule is: third value = first + second.

$(1,1,1)$: $1+1 = 2$, not 1. No.

$(2,-1,1)$: $2 + (-1) = 1$. Yes!

$(2,-1,3)$: 1 , not 3. No.

$(0,2,1)$: $0+2 = 2$, not 1. No.

Answer: (b) $(2, -1, 1)$

Q7. $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$. Compute $2A - B$.

$$2A = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$$

$$2A - B = \begin{bmatrix} 2-4 & -4-1 \\ 6-(-1) & 0-2 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -5 \\ 7 & -2 \end{bmatrix}$$

Answer: (d) $\begin{bmatrix} -2 & -5 \\ 7 & -2 \end{bmatrix}$

Q8. Select the subspace of $\mathbb{R}^2$.

A subspace must contain $(0,0)$ and be closed under $+$ and scaling.

$x + y = 1$: at $(0,0)$ gives $0 = 1$. Fails.

$xy = 0$: $(1,0)$ and $(0,1)$ are in it, but sum $(1,1)$ has $xy = 1$. Fails.

$x - 2y = 0$: $(0,0)$ works; if $x=2y$ and $x'=2y'$ then sum and scalar also satisfy. Passes.

$x \geq 0$: take $(1,0)$, multiply by $-1 \rightarrow (-1,0)$ not in set. Fails.

Answer: (c) $\{ (x,y) : x - 2y = 0 \}$

Q9. Find (a,b) with $a(1,2) + b(2,-1) = (5,0)$.

First coordinate: $a + 2b = 5$

Second coordinate: $2a - b = 0 \Rightarrow b = 2a$

Put in first: $a + 4a = 5 \Rightarrow 5a = 5 \Rightarrow a = 1$

$b = 2(1) = 2$

Check: $1(1,2) + 2(2,-1) = (1+4, 2-2) = (5,0)$. Correct.

Answer: (c) $(1, 2)$

Q10. $\dim \{ x \text{ in } \mathbb{R}^3 \mid x_1 - x_2 = x_3 \}$

One equation, three variables.

Free: $x_2 = s$, $x_3 = t$; then $x_1 = s + t$.

$x = s(1,1,0) + t(1,0,1)$. Two basis vectors.

Rule: $3 - 1 = 2$

Answer: (d) 2

Q11. $\dim \{ x \text{ in } \mathbb{R}^3 \mid x_1 = 0 \}$

Points look like $(0, s, t)$.

Basis: $(0,1,0)$ and $(0,0,1)$.

Rule: $3 - 1 = 2$

Answer: (d) 2

Q12. $\dim \{ x \text{ in } \mathbb{R}^2 \mid x_1 = 0, x_2 \text{ in } \mathbb{R} \}$

Points look like $(0, t)$ - the y-axis.

Basis: $(0,1)$. Only one vector.

Answer: (b) 1

Q13. Find k for infinitely many solutions: $kx + y = 2$ and $2x + 2y = 4$.

Divide second equation by 2: $x + y = 2$.

Infinitely many solutions = same line, so coefficients must match.

$kx + y = 2$ must equal $x + y = 2 \Rightarrow k = 1$

Check with determinant: $\begin{bmatrix} k & 1 \\ 2 & 2 \end{bmatrix} \rightarrow 2k - 2 = 0 \rightarrow k = 1$.

Answer: (d) 1

Q14. $\dim \{ x \text{ in } \mathbb{R}^2 \mid x_2 = 2x_1 \}$

Let $x_1 = t$, then $x_2 = 2t$.

$x = t(1, 2)$. One basis vector.

Rule: $2 - 1 = 1$

Answer: (d) 1

Q15. Which matrix is positive semi-definite? (a) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3 \\ 2 & 8 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$

PSD needs symmetric matrix with all eigenvalues ≥ 0 .

(a) eigenvalues -1 and 1 . Negative one. Fails.

(b) and (d) are not symmetric ($3 \neq 2$). Fails the standard test.

(c) zero matrix: $x^T 0 x = 0 \geq 0$ for every x . Eigenvalues $0, 0$.

So the zero matrix is PSD (but not positive definite).

Answer: (c) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Q16. Which subset of $\mathbb{R}^2$ is a subspace?

$y = 1$: does not contain $(0, 0)$. Fails.

$x + y = 0$: contains $(0, 0)$; sum and scalar multiple keep $x + y = 0$. Passes.

$x \geq 0$: $(1, 0)$ times -1 gives $(-1, 0)$, outside. Fails.

$x - y = 1$: at $(0, 0)$ gives $0 = 1$. Fails.

Answer: (b) $\{(x, y) : x + y = 0\}$

Q17. Which matrix is positive definite? (a) $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3 \\ 2 & 8 \end{bmatrix}$ (c) $\begin{bmatrix} 7 & 0 \\ 0 & 25 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

PD needs symmetric with all eigenvalues > 0 (strictly).

(a) eigenvalues $-1, 0$. Fails.

(b) not symmetric. Fails.

(c) diagonal matrix $\rightarrow$ eigenvalues are 7 and 25 , both > 0 . Passes.

(d) all eigenvalues 0 , so only PSD, not PD.

Answer: (c) $\begin{bmatrix} 7 & 0 \\ 0 & 25 \end{bmatrix}$

Q18. Which matrix is positive semi-definite? (a) $\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$

(a) not symmetric. Fails.

(b) eigenvalues 1 and 0 , both ≥ 0 . Passes.

(c) eigenvalues $-1, -1$. Negative. Fails.

(d) eigenvalue -2 . Fails.

Check (b) directly: $x^T A x = x_1^2 \geq 0$ always.

Answer: (b) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Q19. Solve $2x + y = 7$ and $x - y = 2$.

Add both equations: $(2x + y) + (x - y) = 7 + 2$

$3x = 9 \Rightarrow x = 3$

Put $x = 3$ in $x - y = 2$: $3 - y = 2 \Rightarrow y = 1$

Check: $2(3) + 1 = 7$. Correct.

Answer: (b) $(3, 1)$